

Assessing Sex as a Modifier of Gastric Microbiome Composition and Function Across Histopathological Stages of Gastric Carcinogenesis

Introduction & Background

Introduction to GC

Gastric cancer is considered a major global health challenge. It is the fifth most diagnosed cancer in the world, and due to its frequent late diagnosis, it is the third leading cause of cancer-related deaths (1). Although gastric cancer incidence and associated mortality have decreased in the past decades thanks to medical advancement, an increasing trend has been observed among younger populations (2). This highlights the importance of research in determining the risk factors associated with the development and progression of gastric cancer. This disease follows a four-stage progression timeline known as the Correa Cascade. The first step involves the presence of chronic inflammation (chronic gastritis) in the stomach that may progress to intestinal metaplasia (IM), where gastric epithelium is replaced by intestinal epithelium (3). These lesions may then advance to intraepithelial neoplasia (IN) or dysplasia, which represents the phenotypic bridge between premalignant tissue changes and invasive cancer (4). Intraepithelial neoplasia (IN) is particularly important as it is often treated as a phenotypic bridge between metaplastic lesions and invasive cancer (5). Many risk factors have been associated with the incidence of gastric cancer, including diet, smoking, alcohol, genetic predisposition, infections, and other medical conditions (6). Of the stated factors, the human microbiome has also been recently linked to a variety of medical conditions, including gastric cancer.

Microbiomes role's + gap

The human microbiome refers to the grouping of bacteria, archaea, viruses and eukaryotes found within and outside of our body (grace a). These organisms impact different parts of human physiology, positively or negatively, via enhancing or impairing metabolic and immune functions. The stomach – despite its acidic properties – contains several bacterial phyla, including *Proteobacteria*, *Bacteroidetes*, *Fusobacteria*, *Actinobacteria*, and *Firmicutes*, thus it is not surprising that this bacterial community has been linked to gastric cancer (lili wang). For example, a major carcinogen associated with the development of gastric cancer is *Helicobacter pylori*, whose presence often results in erosion of gastric epithelium as a result of chronic inflammation. The progression from gastritis to gastric cancer is also associated with diversification of the microbiota and enrichment of bacterial genera with cancer-promoting profiles (lili wang). Though the gastric microbiome is associated with cancer, its ability to promote carcinogenesis is modulated by many host factors, such as biological sex, highlighting the need for more in-depth analysis.

The main paper + ujemi

In one study, researchers Wang et al. examined the changes in the gastric microbiota across different stages of the Correa Cascade using 16s rRNA sequencing of provided gastric mucosal biopsy samples. A total of 132 patients of Han nationality from northern China at different stages of gastric cancer were

enrolled in this study. The study includes 30 healthy controls (HC), 21 chronic gastritis (CG), 27 IM, 25 IN, and 29 gastric cancer (GC) patients. Among the patients, there are 126 females and 184 males. The biopsy samples were subjected to DNA extraction, PCR, sequencing, data and statistical analysis (Wang et al.). The study found that changes in gastric microbiota composition and associated functional profile were strongly associated with stages of gastric carcinogenesis. Using microbial taxa as predictors, they were able to construct a random forest model that accurately predicts histological types. However, the researchers did not investigate the underlying mechanism, nor did they take into account sex-associated differences in the gastric microbiome. A UJEMI study expanded on Wang et al.'s dataset by investigating whether change in age reflects the composition of the gastric microbiota across different stages (Mehra et al.). Although the study's findings suggest that age does not impact overall gastric microbiome diversity, the study highlights the value of testing host-intrinsic factors' modification on patterns of microbiome during gastric cancer progression. In a parallel focus, our study will use the provided dataset by Wang et al. and will expand upon the results by investigating the role of biological sex in the composition of the gastric microbiome across the stages of gastric cancer.

Current work + importance of sex + relay of gap

Across multiple studies, gastric cancer progression is associated with gastric mucosal microbiome dysbiosis (Coker et al. 2018; Wang et al. 2020; Gunathilake et al. 2019). Here, dysbiosis describes disruptions in microbial diversity and community composition compared to healthy controls across different disease stages. Importantly, the specific pattern of dysbiosis may not be uniform across all individuals, because host factors like biological sex can shape microbial community structure. The role of biological sex as a factor in microbiota composition has been gaining popularity over time as a result of increasing evidence linking bacterial communities to the pathogenesis of different diseases. For example, a study found that the abundance of bacterial genera differs in men and women with metabolic syndrome, suggesting an explanation for the difference in prevalence of the condition between the two sexes (Jose et al.). Another study found that Autoimmune Atrophic Gastritis resulted in more severe sex-associated dysbiosis in men, which may explain why gastric cancer is more common in men (Giulia et al.). These studies suggest that males and females may experience different host selection pressures at the mucosal surface. Such sex-associated differences in gut microbiota composition impact the functional profile of the community, which has a downstream impact on host physiology. Given the links between the gut microbiome, host health, biological sex, and disease pathogenesis, exploring the impact of sex on microbiome composition and its changes throughout different stages of gastric cancer may provide insight into sex-dependent mechanisms of cancer progression and targets for therapeutics.

Research Objective and Hypothesis

Research question: Does biological sex modify the relationship between gastric disease histopathological stage and gastric microbiome composition and function?

Prior studies have consistently established that the gastric mucosal microbiome is a critical mediator of gastric cancer progression. Notably, Wang et al. (2020) demonstrated that patients transition through distinct microbial profiles during different stages of the Correa Cascade. These shifts result in significant variations in taxonomic diversity and functional profiles, particularly evident at the "bridge" of intraepithelial neoplasia. Such dysbiosis, characterized by the loss of beneficial commensals and

enrichment of pro-inflammatory taxa, is known to drive carcinogenesis. Along with the overall trend, there is emerging evidence that host-intrinsic factors can potentially modulate these microbial compositions and patterns. Given that biological sex is a potential driver of mucosal immunity and physiological selection pressures, we hypothesize that sex acts as a significant effect modifier of the gastric microbiome during cancer progression. We predict that males and females will display divergent alpha and beta diversity metrics, reflecting sex-specific microbial community complexity and structure. Furthermore, we anticipate identifying sex-dependent "indicator" taxa and altered metabolic pathways that are potentially linked to differential inflammatory responses that are more prominent in one sex than the other.

The rationale for this prediction stems from the fact that biological sex influences the mucosal environment, which in turn selects for specific microbial populations. For instance, sex-associated differences in the abundance of bacterial genera have been linked to the pathogenesis of metabolic syndrome and autoimmune atrophic gastritis, the latter of which shows more severe dysbiosis in men (Pivetta et al., 2023; San-Millán et al., 2020). Since these differences in composition directly impact the functional profile of the community and downstream host physiology, it is probable that the gastric mucosal microbiome follows a sex-stratified trajectory during the transition to malignancy. To investigate these relationships, our study will utilize the 16S rRNA gene sequencing data provided by Wang et al. (2020) and stratify the patient sample by biological sex ("Gender") across all five histopathological stages ("Group"). By employing comparative diversity metrics and function analysis, we aim to find sex-specific taxa that may be currently masked in aggregated groups. This approach will allow us to determine if cancer progression includes a sex-specific mechanism, potentially revealing new, sex-stratified targets for stage prediction and therapeutics.

Experimental aims and rationale

Aim 1 - To determine whether males and females differ in microbial composition in their gastric mucosal microbiome at different progression of histopathological gastric cancer stages

Rough Draft:

Previous work by Wang et al. demonstrated significant changes in gastric microbiome diversity across histopathological stages of gastric disease. Their study performed pairwise comparisons following ANOVA and found that healthy controls (HC), chronic gastritis (CG), and intestinal metaplasia (IM) did not significantly differ from each other. However, as gastric disease progresses, they found that gastric microbiome alpha diversity significantly decreases as seen in the intraepithelial neoplasia (IN) and gastric cancer (GC) groups. Although these findings demonstrate a significant decrease in alpha diversity across later disease stages, they do not explore the mechanisms or factors that may give rise to these changes. Therefore, this analysis investigates whether there is a difference in within-sample gastric mucosal microbial diversity between males and females within each gastric disease histopathological stage (HC, CG, IM, IN, GC). Specifically, this analysis examines whether males and females within the same disease stage differ in the number of unique microbes present (richness), the uniformity of the organisms/ASVs (evenness), and/or how closely related they are to one another phylogenetically (phylogenetic relatedness). By evaluating multiple diversity metrics such as Shannon, Chao1, Simpson, Observed

Features, and Faith's Phylogenetic Diversity, different aspects of the microbial community structure can be captured. A Wilcoxon test will be used to determine whether observed differences are statistically significant using adjusted p-values. If significant differences are observed, this would suggest that biological sex influences the complexity of the gastric mucosal microbiome during gastric disease progression. However, if no differences are observed (non-significant results), this would suggest that sex does not drive changes in microbial diversity, and that other factors may be responsible.

By assessing beta diversity among different samples from their mucosal gastric microbiome, we can determine whether microbial composition differs between sexes. This comparison allows us to evaluate whether biological sex is associated with changes in gastric mucosal microbial composition as gastric disease progresses. If the analysis detects a difference, it may suggest that sex-related factors influence the change in the gastric mucosal microbiome during gastric disease progression. Furthermore, understanding microbiome changes that contribute to the progression of gastric disease, may help us identify ways to prevent the disease from developing further before the patient's condition becomes critical.

Aim 2 - To determine whether males and females differ in the taxonomic composition of the gastric mucosal microbiome and microbial indicator taxa across histopathological stages of gastric cancer progression.

Building on the foundation laid by Wang et al. (2020), we know that the gastric microbiome undergoes a systematic shift as it progresses through the stages of carcinogenesis from healthy tissue to chronic gastritis, and eventually to gastric cancer. Their work highlighted a critical transition at the intraepithelial neoplasia (IN) stage, where microbial signatures shift toward an enrichment of taxa like *Streptococcus* and *Fusobacterium*. While these markers exhibit a clear progression, it did not directly address whether sex modifies which microbes best characterize each stage. Because biological sex can influence mucosal immunity and inflammatory response (Sankaran-Walters et al., 2013), the same histological stage may not correspond to the same indicator microbial taxa in males compared to females. By applying indicator taxa analysis to sex-stratified samples, we can determine if these stage-linked indicator microbes are shared across both sexes or if they are sex-specific. Through this approach, we can identify which microbes are highly enriched and specific to each sex across disease stages. If certain taxa emerge as indicators in only one sex, it would suggest that biological sex can potentially act as a modifier of the microbiome's role in cancer progression. These sex-dependent indicator taxa may serve as biomarkers of gastric disease progression. It may provide potential for sex-stratified prediction of histopathological stage and improving outcome classification separately in males and females.

Aim 3 - To determine whether biological sex influences the functional profile of the gastric mucosal microbiome across different stages of gastric carcinogenesis

As identified by Wang et al. (2020) in their paper "Changes of the Gastric Mucosal Microbiome Associated With Histological Stages of Gastric Carcinogenesis", the gastric mucosal microbiome is not static during gastric carcinogenesis. This was especially evident when gastric cancer progressed from intestinal metaplasia (IM) to dysplasia (IN) and ultimately cancer (GC), with many pathways related to

bacterial toxin expression and DNA repair being upregulated. Although these findings suggest that a change in microbiome composition is accompanied by shifts in functionality during the later stages of carcinogenesis, the authors did not consider the impact of biological sex. This factor should be included when discussing the pathogenesis of disease, as the male and female bodies differ in their ability to mount immune responses (Wang et al., 2019). In addition, the immune system is regulated by the gastric microbiota, and studying the influence of sex on the function profile of the gastric microbiome may help explain why gastric cancer is more prevalent in males (Wiertsema, Selma P et al., 2021; Ma, Junfu et al., 2022). By applying functional analysis to sex-stratified samples, it is possible to determine if biological sex plays a role in carcinogenesis via the regulation of different pathways. This analysis will uncover which – if any – pathways are upregulated or downregulated across these stages between the sexes. A combination of this data with literature could unveil the underlying mechanism behind gastric carcinogenesis. Furthermore, this analysis could pave the way for new sex-specific therapeutics that may be more effective than current options.

Proposed approach

Data Processing					
1. In Qiime2, import sequences in the form of a manifest file and generate a demux.qza file for further analysis 2. Denoising with DADA2 (truncation length of 252 nt) and clustering to generate table.qza and rep-seqs.qza file 3. Training and applying of classifier for taxonomic assignment 4. Filtering for non-bacterial sequences such as mitochondria and chloroplasts 5. Filtering for ASVs that have assignments at least to the phylum level 6. Generation of phylogenetic tree and alpha rarefaction curve to determine the sampling depth that will be used for downstream analysis 7. Export of required files (metadata.tsv, feature-table.txt, tree.nwk, taxonomy.tsv) from QIIME2 8. In R, install the required packages and load the files exported from QIIME2 9. Modify the imported files to the appropriate format, convert the feature-table and taxonomy to matrices 10. Production of phyloseq object using exported files					

	Aim 1		Aim 2: Across gastric disease histopathological stages, do male and female patients show different taxonomic composition patterns and distinct stage-specific microbial indicator taxa?		Aim 3: Does biological sex influence the functional potential of the gastric mucosal microbiome during gastric carcinogenesis?
Do you plan to use			Indicator taxa Analysis No	Differential Abundance Yes. - All samples	Yes. - All samples are included in the comparison but will be filtered depending on the

every single sample in this aim? (yes/no)			- Filter to keep ASVs with a relative abundance greater than 0.001	are included, however taxa are filtered using a 0.1 prevalence cutoff (taxon must occur in $\geq 10\%$ of samples within the given stage)	analysis (e.g., functional analysis of male and female in the IN stage would filter the Group column to include only that stage). - Graph generation will filter data to include only significant results (p-adjust > 0.05 and log2fold change of 1> or <-1).	
What major analyses/analyses will you run? (ex. alpha diversity)			Indicator taxa analysis with multipatt() : 1. Aggregate the ASVs to the genus level 2. Convert to relative abundance + Filter 3. Run Indicator Taxa analysis stratified by "Group" 4. Production of publication-ready graphs with the top 10 statistically significant indicator genus for each gastric disease stage 5. Create a separate phyloseq object for each gastric disease stage ("Group") 6. Run	Differential abundance analysis with ANCOM-BC: 1. Aggregate the ASVs to the genus level 2. Set female as the reference level 3. Stratify by "Group" and test differential abundance by "Gender" within each stage 4. Production of publication-ready plots of significantly different taxa between males and females within each stage	1. Filtering by gastric cancer stage ("Group") in R 2. Differential analysis and annotation comparing stages by biological sex 3. Production of publication-ready graphs with significant up/downregulated pathways using ggppirust in R	1. Filtering by biological sex ("Gender") in R 2. Differential analysis and annotation comparing stages to healthy control (HC) 3. Production of publication-ready graphs with significant up/downregulated pathways using ggppirust in R
					4. Generation of a heat map for supplemental analysis 5. Comparison of results with the literature and noting down of matching and conflicting observations	

			indicator taxa analysis stratified by sex at each stage 7. Production of publication-ready graphs with the top 10 statistically significant indicator genus for each sex at each disease stage		
What statistical tests will you use for each major analysis?			- permutation test of taxon-group association (within multipatt()) - Significance cutoff: p.value <0.05 and stat >0.7	- ANCOM-BC differential abundance test of taxon-sex association (within each histopathological stage) - Significance cutoff: BH-adjusted p-value (< 0.05) - log2 fold change used to report effect size (male relative to female)	- LinDA will be used to calculate differential abundance - Significance cutoff: BH-adjusted p-value > 0.05 and log2 fold change of 1 > or < -1
How will you visualize your results?			Generation of dotplots to show different indicator taxa between groups of interest - Between gastric disease	Generation of barplots of log2 fold changes to display the significantly different genera between males and females within each gastric disease stage	Generation of a graph displaying the functional difference between samples (name of pathway affected, metabolite affected, upregulation or downregulation of affected pathway) Generation of a heatmap to identify similar and/or

			stages - Between sex → separate plot for each gastric disease stage - Boxplots for close-up comparison of sex-specific genus.		different pathways Generation of PCA plot to determine which samples are more similar to each other
What key programs (i.e. QIIME 2, R) and key R packages will you require?	R	R	R Packages: 1. Tidyverse 2. Phyloseq 3. Indicspecies	R Packages: 1. Tidyverse 2. Phyloseq 3. ANCOMBC	QIIME2 1. PICRUST2 environment R Packages 1. Tidyverse 2. Phyloseq 3. Ggpicrust2 4. “pheatmap”

Participant report

Teammate	Proposal Contribution	Analysis
Christina Xugao	- Introduction: Dataset description - Proposed approach: Aim 2 - Indicator taxa - Research Objective and Hypothesis	Indicator Taxa Analysis Core microbiome
Ryan Wong	- Aim 2 - Differential Abundance - References	Differential Abundance Analysis
Parsa Shafiekhani	- Introduction - draft introduction paragraphs - Aim 3 - Functional Analysis	Functional Analysis
Saray Fallahnafari	- Aim 1	

	- References	
Amitis Eskandari	- Aim 1	

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Reference Links (keep in this format for now, i will organize later)

<https://pubmed.ncbi.nlm.nih.gov/30207593/>

<https://link.springer.com/article/10.1186/s12889-024-19104-6>

<https://www.frontiersin.org/journals/cell-and-developmental-biology/articles/10.3389/fcell.2025.1620756/full>

<https://www.sciencedirect.com/science/article/abs/pii/S1521691813000735>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4739309/> (big paper, i think)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5980829/>

<https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2020.00997/full>

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Team 1

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